

Definitions

As before, we have the following quantities:

S = The set that contains all legal chess positions as a subset
 N = The size of S ($2.19645e53$)
 S_{sample} = A random sample from S
 N_1 = The size of S_{sample} ($2.746e15$)
 $S_{n1_candidates}$ = The computed subset of S_{sample} containing
all potentially legal positions
 $N_{n1_candidates}$ = The size of $S_{n1_candidates}$ (6030640)
 S_2 = A random sample from $S_{n1_candidates}$
 N_2 = The size of S_2 ($1e5$)
 N_{s2_legal} = The number of legal positions in S_2 (99983)
 $N_{s2_illegal}$ = The number of illegal positions in S_2 (17)
 S_{n1_legal} = The set of legal positions in S_{sample}
 N_{n1_legal} = The size of S_{n1_legal} (not computed exactly)
 $N_{n1_legal_est}$ = Estimated number of legal positions
in S_{sample} (6029615)
 $N_{\text{legal_est}}$ = Estimated number of legal chess positions
($4.823692e44$)

We introduce the following unknown parameters:

- N_{legal} = The number of positions in S that are legal chess positions. S_{legal} is the corresponding set. This is the main unknown parameter to be estimated, $\approx 4.8 \cdot 10^{44}$.
- $N_{\text{candidates}}$ = The number of positions in S that are potentially legal according to the first automatic analysis. $S_{\text{candidates}}$ is the corresponding set. This is the nuisance parameter, $\approx 4.8 \cdot 10^{44}$, $> N_{\text{legal}}$.

Introduce the following variables:

$$n = N_1 \tag{1}$$

$$m = N_2 \tag{2}$$

$$C = N_{n1_candidates} \tag{3}$$

$$L = N_{n1_legal} \tag{4}$$

$$H = N_{s2_legal} \tag{5}$$

$$p = \frac{N_{\text{candidates}}}{N} \tag{6}$$

$$q = \frac{N_{\text{legal}}}{N_{\text{candidates}}} \tag{7}$$

That is, n is the size of the first sample, m is the size of the second sample, p is the fraction of positions in S that are potentially legal and q is the fraction of positions in $S_{\text{candidates}}$ that are legal.

Since p can be estimated by $\frac{C}{n}$ and q by $\frac{H}{m}$, the estimate $\hat{N}_{\text{legal}}$ of the number of legal chess positions is:

$$\hat{N}_{\text{legal}} = N \frac{C}{n} \frac{H}{m} \tag{8}$$

Expected value

Because of the way the sampling procedure is defined, C , L and H are random variables with the following distributions:

$$C \sim \text{Binomial}(n, p) \quad (9)$$

$$L \mid C \sim \text{Binomial}(C, q) \quad (10)$$

$$H \mid C, L \sim \text{HyperGeometric}(C, L, m) \quad (11)$$

For the distributions, we have [1, 2]:

$$\mathbb{E}[C] = np \quad (12)$$

$$\mathbb{E}[L \mid C] = Cq \quad (13)$$

$$\mathbb{E}[H \mid C, L] = m \frac{L}{C} \quad (14)$$

so

$$\mathbb{E}[\hat{N}_{\text{legal}} \mid C, L] = \mathbb{E}\left[N \frac{C}{n} \frac{H}{m} \mid C, L\right] \quad (15)$$

$$= \frac{N}{n} \frac{C}{m} \mathbb{E}[H \mid C, L] \quad (16)$$

$$= \frac{N}{n} \frac{C}{m} m \frac{L}{C} = \frac{N}{n} L \quad (17)$$

Taking expectations again and using the law of iterated expectations [3],

$$\mathbb{E}[\hat{N}_{\text{legal}} \mid C] = \frac{N}{n} \mathbb{E}[L \mid C] = \frac{N}{n} Cq \quad (18)$$

Taking expectations once more,

$$\mathbb{E}[\hat{N}_{\text{legal}}] = \frac{N}{n} q \mathbb{E}[C] = \frac{N}{n} q np = Npq \quad (19)$$

$$= N \frac{N_{\text{candidates}}}{N} \frac{N_{\text{legal}}}{N_{\text{candidates}}} \quad (20)$$

$$= N_{\text{legal}} \quad (21)$$

Therefore the estimator is unbiased.

Variance

The variance of a product of two independent random variables X , Y , is [4]:

$$\text{Var}(XY) = \mathbb{E}[X]^2 \text{Var}(Y) + \mathbb{E}[Y]^2 \text{Var}(X) + \text{Var}(X) \text{Var}(Y) \quad (22)$$

which can be written

$$\frac{\text{Var}(XY)}{\mathbb{E}[XY]^2} = a + b + ab \quad (23)$$

where

$$a = \frac{\text{Var}(X)}{\mathbb{E}[X]^2}, \quad b = \frac{\text{Var}(Y)}{\mathbb{E}[Y]^2} \quad (24)$$

The estimator is

$$\hat{N}_{\text{legal}} = N \frac{C}{n} \frac{H}{m} \quad (25)$$

Let

$$X = \frac{C}{n}, \quad Y = \frac{H}{m} \quad (26)$$

Although H is obtained by sampling from the C candidates, the legality of each candidate is an independent Bernoulli [5] random variable with probability q , so¹

$$H \mid C \sim \text{Binomial}(m, q) \quad (27)$$

which does not depend on C . Since n and m are just constants, this means that X and Y are also independent, so 23 and 24 are applicable. We have

$$a = \frac{np(1-p)}{(np)^2} = \frac{1-p}{np} \quad (28)$$

$$b = \frac{mq(1-q)}{(mq)^2} = \frac{1-q}{mq} \quad (29)$$

Using 25, 26, 23, 21

$$\text{Var}(\hat{N}_{\text{legal}}) = \text{Var}(NXY) = N^2 \text{Var}(XY) \quad (30)$$

$$= N^2 \mathbb{E}[XY]^2 (a + b + ab) = \mathbb{E}[NXY]^2 (a + b + ab) \quad (31)$$

$$= \mathbb{E}[\hat{N}_{\text{legal}}]^2 (a + b + ab) \quad (32)$$

$$= N_{\text{legal}}^2 (a + b + ab) \quad (33)$$

Finally, taking square roots:

$$\text{SD}(\hat{N}_{\text{legal}}) = N_{\text{legal}} \sqrt{a + b + ab} \quad (34)$$

¹Assuming $C \geq m$. For sample sizes of interest, $\Pr(C < m)$ is extremely small. If $C < m$, all C candidates can instead be analyzed and the estimator $\frac{NL}{n}$ used, with value zero when $C = 0$. This preserves unbiasedness, but the variance formula 34 does not cover that fallback.

References

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